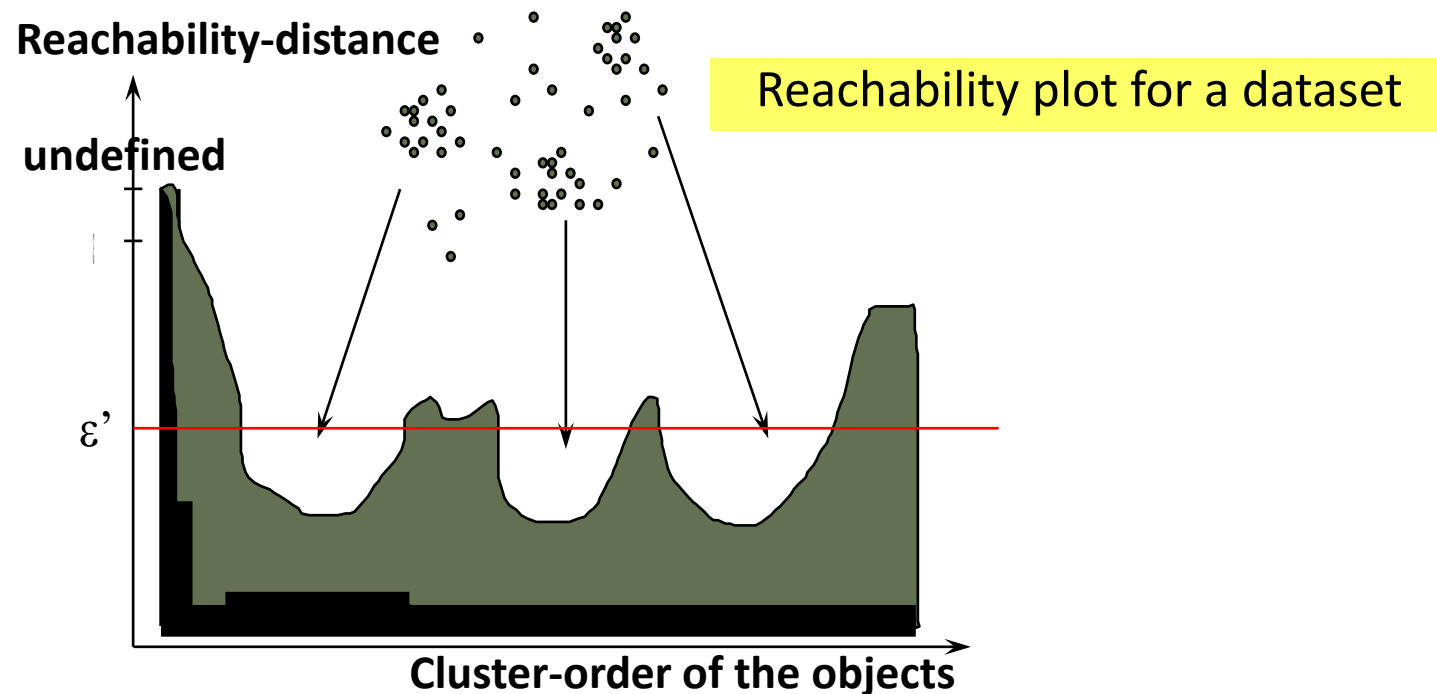


Visualization

- ❑ Since points belonging to a cluster have a low reachability distance to their nearest neighbor, valleys correspond to clusters
- ❑ The deeper the valley, the denser the cluster



OPTICS: An Extension from DBSCAN

- Core distance of an object p : The smallest value ϵ such that the ϵ -neighborhood of p has at least MinPts objects
- Let $N_\epsilon(p)$: ϵ -neighborhood of p , where ϵ is a distance value
- Core-distance $_{\epsilon, \text{MinPts}}(p)$ = Undefined if $\text{card}(N_\epsilon(p)) < \text{MinPts}$;
Core-distance $_{\epsilon, \text{MinPts}}(p)$ = MinPts-distance(p), otherwise

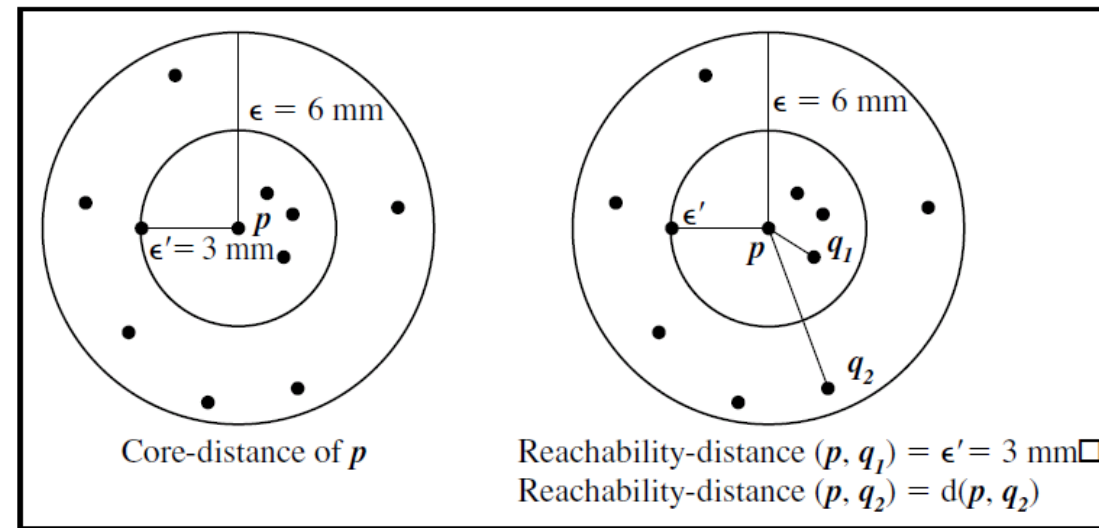
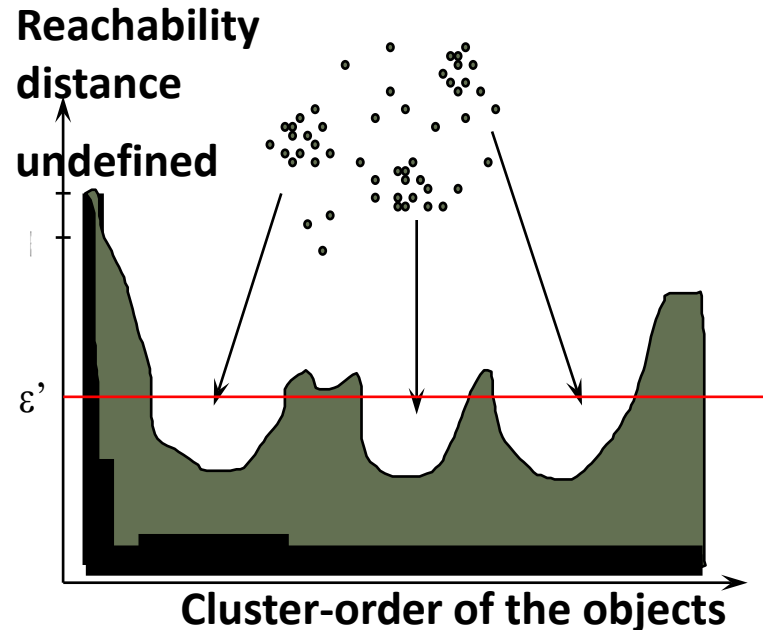


Figure 10.16: OPTICS terminology. Based on [ABKS99].

OPTICS: An Extension from DBSCAN

- Reachability distance of object q from core object p is the minimum radius value that makes q density-reachable from p

Reachability-distance $_{\epsilon, MinPts}(p, q)$ = Undefined, *if p is not a core object*

Reachability-distance $_{\epsilon, MinPts}(p, q)$ = $\max(\text{core-distance}(p), \text{distance}(p, q))$, *otherwise*

- Complexity: $O(N \log N)$ (if index-based), where N : # of points